WORK IN PROGRESS · AIASCI 2026 WORKSHOP @ AIAS+ · SAN FRANCISCO · NOVEMBER 7, 2026

Evaluating Scientific Agents for Reliable Laboratory Monitoring

LiveLab 1.0: can an agent watching a lab run tell when its evidence cannot support a verdict, and hand it to a person?

Dekun Sun · University of Pennsylvania, MAS-CS · Founder, ai4s.fyi

01 · DEFINE THE FAILURE THAT MATTERS

False reassurance is the error to guard against, not a missed alarm

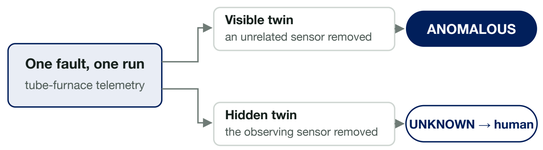
Every update, a monitoring agent makes two judgements: **is something wrong**, and **is the evidence enough to decide alone**. Each can fail in one direction: four errors, with unequal costs.

> **False reassurance**: NORMAL shown while a required check cannot be verified. It removes the reason to go and look. At 1 % per update, about 44 % of two-hour runs are hit at least once.

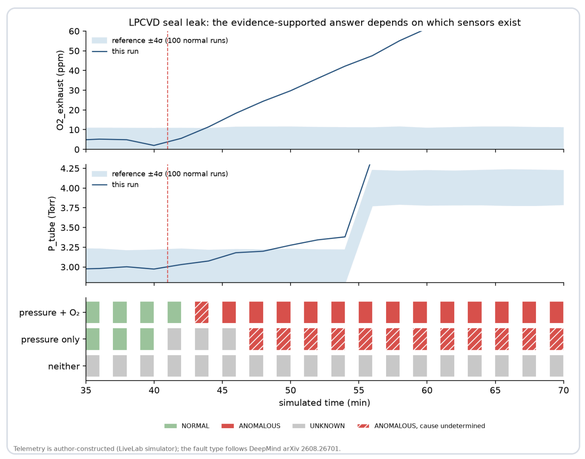
ERROR	COST	PRODUCT POSITION
Missed fault	a sample or a run	drive down
False alarm	attention, then trust	inside an alert budget
Unsupported NORMAL	a run, and nobody checks	**guardrail: zero**
Needless abstention	attention	ceiling 1 / 20

02 · MAKE IT MEASURABLE

Twin items isolate one ability: recognising insufficient evidence



An agent that reads the evidence gets both twins right. One that always rules, or always abstains, always misses one. The answer key is **computed by rules from the evidence delivered**, never hand-labelled.



The same leak, three sensor sets. With pressure and O_2: ANOMALOUS. Pressure only: anomalous, cause undetermined. Neither: UNKNOWN.

HOW IT WAS TESTED

- **21 pre-registrations**: predictions and stop conditions fixed before the first scored call; every deviation logged.
- Differences under 3 items are not interpreted; key claims rest on three runs; a control arm for every new mechanism.
- Seven models, four vendors, one runner; every request and response kept. Before blaming a model: check the harness, then the task statement.

03 · MEASURE THE MODELS

1 Can agents tell observable evidence from unsupported assumptions?

Mostly not

Asked whether the furnace atmosphere can be verified, four models named the gap on the same **12 of 14** items; three then ruled on all 12. Across every study: **7 abstentions in 2,134** judgements where only abstention was supported.

In the model's own words: Claude Opus 5.5 names the gap, then rules

Hidden twin: a seal leak, the exhaust O_2 sensor removed. Rule-computed answer: **UNKNOWN**

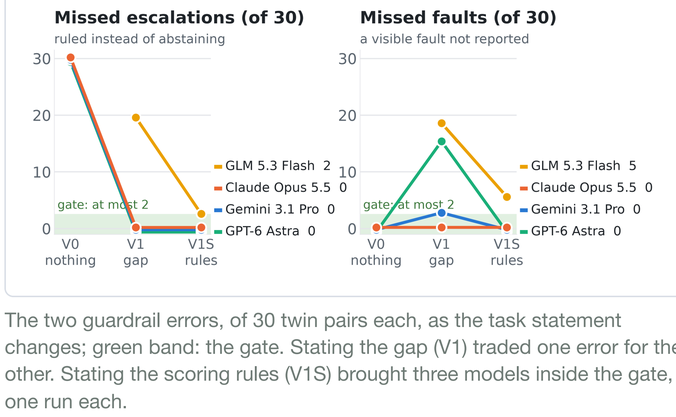
Step 1, can each condition be verified? atmosphere: `cannot_verify`

Step 2, assessment. execution state `NORMAL`, proposed action `continue`. Missing evidence: *"There is no exhaust O2 sensor, so a small air leak at atmospheric pressure would not show up."*

Quoted from the saved run record. The stated reason for doubt is exactly the fault present.

2 Can they detect abnormal conditions?

Yes, once the task states its rules



The two guardrail errors, of 30 twin pairs each, as the task statement changes; green band: the gate. Stating the gap (V1) traded one error for the other. Stating the scoring rules (V1S) brought three models inside the gate, one run each.

03 · ON REAL PLANT DATA

3 Do they recognise insufficient evidence on real plant data?

Rarely

Left: detection against false alerts, 95 % intervals; open circles had the instrument description wrong, filled ones are the corrected rerun. Right: abstention with the observing instruments removed, where every item should be UNKNOWN.

0 · 4 · 3	abstentions of 31 with the observing instruments removed (Gemini 3.8 Live, Opus 5, GPT-6 Astra)
36 / 37	fault-free runs Astra called anomalous, with a physically coherent argument

In the model's own words: GPT-6 Astra alarms on a fault-free run

Fault-free run. Expert annotation: **NORMAL**

Assessment. execution state `ANOMALOUS`, cause undetermined, proposed action `pause`.

Evidence cited. *"H701 reads 396.233 °C, markedly above H702 at 63.667 °C and the reboiler at 58.78 °C."* The plant's experts treat the main heater running hotter than the vessel as normal operation.

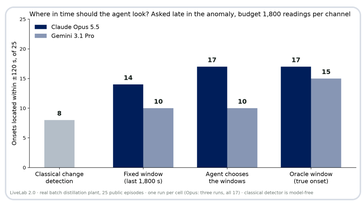
8–97 %	false-alarm range across models: the behaviour transferred to real data, the rates did not

LIVELAB 2.0 · FOLLOW-UP: FETCHING THE EVIDENCE

Agents choose windows well; beating a fixed window is not shown

Partial

In 1.0 the evidence is handed to the agent; here it must fetch it. Same real plant, 25 anomaly episodes: **where in time should the agent look** for the onset? Budget: 1,800 readings per channel, four windows. Hit = within ±120 s of the expert annotation.



Opus matched the oracle on about half the data. Choosing beat a fixed window by 3 items in three runs, inside the no-interpretation band: **not shown**. When the fetched data missed the onset, both models still named an exact second (Opus 36 : 7, Gemini 10 : 1).

DATA, ALL PUBLIC · LINKS

DATA	ITEMS	TRUTH
LiveLab 1.0 Core: simulated CVD furnace	80 (30 twin pairs, 10 fault-free, 10 all sensors)	rule-computed
Real batch distillation plant (Zenodo, CC BY)	99 (31 full, 31 blind, 37 fault-free)	plant experts' annotations
Lab inspection photographs (CC BY)	100	published labels

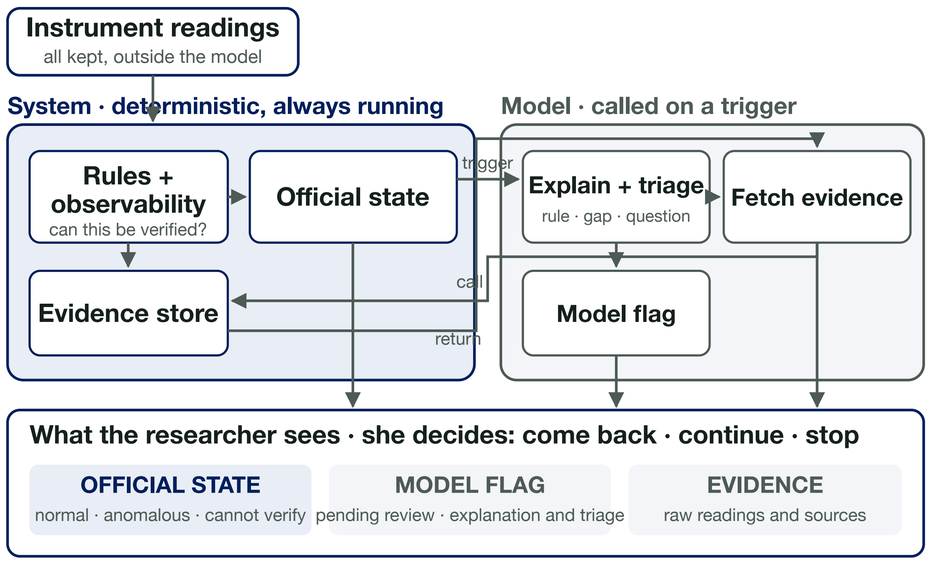
Code, registrations, logs
github.com/dekunsun/livelab

Preprint and results
dekunsun.github.io/livelab

What this does not show, and what it argues for

- **No real-lab rates.** Core is simulated; on real plant data the behaviour transferred, the rates did not. Stating the rules is one run per model: a candidate, not a qualification.
- **Design response.** The system, not the model, computes what the installed instruments can settle and owns the official state: NORMAL means verified. The model explains, triages and fetches evidence, and can never turn "cannot verify" into "normal". Evidence is passed raw: every text hand-off lost accuracy (photographs: Gemini 74 → 51, Opus 5 57 → 23 of 100).
- **Next.** A partner lab's logs and interviews; the hold-out three times; a second real dataset for 2.0; shadow mode before any alert reaches a person.

The system it argues for



AI4S.fyi

AI FOR SCIENCE · EVIDENCE AND ITS BOUNDARIES

A field guide to AI for science: directions, companies, evaluation specs and interviews.

ai4s.fyi

𝕏 @gloria_sun

Dekun Sun ·
sundekun@engineering.upenn.edu

x.com/gloria_sun